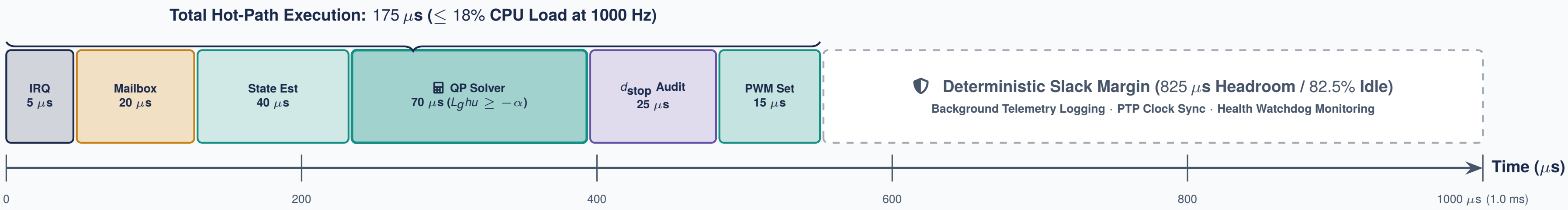
1000 HZ BARE-METAL MCU TIMING TICK & ZERO-ALLOCATION BUDGET

Deterministic Microsecond Execution on ARM Cortex-M4 Silicon vs. Linux MPU Scheduler Latency Jitter

System 1 Real-Time Domain: Cortex-M4 Hard Real-Time 1.0 ms (1000 μs) ISR Slot

Guarantees: Zero Dynamic Heap Allocations (malloc = 0 B), TCM Static SRAM Buffers, Memory-Mapped Timer Registers.



The Linux MPU Reality (System 2): Non-Deterministic Scheduling Jitter

Why learned models running on application processors cannot hold direct electrical gate authority.

The Architectural Invariant: **System 1 (MCU):** Hard 1000 Hz tick guarantees safety within $< 200\,\mu$s, completely independent of Linux load. **System 2 (MPU):** Emits candidate action chunks asynchronously; late or stale proposals are safely ignored.